

Exercise 1

1. What is meant by classification? Explain the two phases of classification.
2. Please provide a real-world example of a classification problem.
3. What is the learning set? Why is this set split into training and test sets?
4. What is a target attribute? What properties must it have to be used for classification?
5. Please describe the variable type of each variables in Table 1. [numerical, categorical, ordinal]

Table 1: Training set for defaulted borrower classification.

Home Owner	Marital Status	Level of Education	Annual Income	Defaulted Borrower
Y	Single	High	125	N
N	Married	Moderate	100	N
N	Single	Low	70	N
Y	Married	High	120	N
N	Divorced	Low	95	Y
N	Married	Moderate	60	N
Y	Divorced	High	220	N
N	Single	Low	85	Y
N	Married	Moderate	75	N
N	Single	High	90	Y

Exercise 2

1. A patient goes to see a doctor. Suppose the doctor performs a test with 99% reliability, i.e., 99% of people who are sick test positive and 99% of the healthy people test negative. The doctor knows that only 1% of the people in the country are sick. If a patient has tested positive, what is the probability he is sick?
2. How does a Naïve Bayes classifier work? Explain using the underlying formula. Why is it “naïve”?
3. How does the Naïve Bayes classifier handle numeric attributes? Give the formula.
4. For the dataset provided in Table 1, compute the conditional probabilities of all attribute values, given the target attribute Defaulted Borrower (DB=Y and DB=N). Classify the following instance according to the the Naïve Bayes Classifier without handling zero conditional probabilities:

$$X = (\text{Home Owner} = N, \text{Marital Status} = \text{Married}, \text{Annual Income} = 90)$$

5. Explain the zero-frequency problem. How can it be avoided?

For the dataset provided in Table 1, calculate the conditional probabilities of all attribute, given the target attribute Defaulted Borrower (DB=Y and DB=N) using Laplace estimate and then classify the following instance:

$$X = (\text{Home Owner} = N, \text{Marital Status} = \text{Married}, \text{Annual Income} = 90)$$

Table 1: Training set for defaulted borrower classification.

Home Owner	Marital Status	Annual Income	Defaulted Borrower
Y	Single	125	N
N	Married	100	N
N	Single	70	N
Y	Married	120	N
N	Divorced	95	Y
N	Married	60	N
Y	Divorced	220	N
N	Single	85	Y
N	Married	75	N
N	Single	90	Y

Exercise 3

1. Describe a potential problem of using **Information Gain** as a splitting criterion in decision trees. Explain why the **Gain Ratio** measure is a way to overcome this problem.
2. Consider the training examples shown in Table 1 for a binary classification problem. Calculate **Information Gain** and **Gain Ratio** for each of the attributes *ID*, *Gender*, *Graduation* and *Eye Color*. Use multiway splits when possible.
3. There exist several approaches to measure the generalization error rate of a decision tree T . The simplest measure is termed as **optimistic error estimate** $\text{err}_o(T)$, which is simply the error rate of T on the training set. In contrast, the **pessimistic error estimate** $\text{err}_p(T)$ also incorporates the “complexity” of T , quantified by the number of leaf nodes.

Let k be the number of leaf nodes and N_{train} be the number of training instances. The pessimistic error estimate is computed as:

$$\text{err}_p(T) = \text{err}_o(T) + \Omega \cdot \frac{k}{N_{\text{train}}}$$

where Ω is a hyperparameter that makes a trade-off between minimizing training error and minimizing model complexity.

A third generalization error estimate is **reduced error pruning** (err_{rep}), which is the error rate of T on a validation set.

Calculate $\text{err}_o(T)$, $\text{err}_p(T)$, and $\text{err}_{\text{rep}}(T)$ for each of the two decision trees in Figure 1. Assume that each leaf node is labeled according to the majority class of training instances that reach the node. When calculating the pessimistic error estimate, assume that $\Omega = 2$.

Reduced error pruning steps:

- Consider each node for pruning.
- *Pruning* = removing the subtree at that node, make it a leaf and assign the most common class at that node.
- A node is removed if the resulting tree performs no worse than the original on the validation set — removes coincidences and errors.
- Nodes are removed iteratively choosing the node whose removal most increases the decision tree accuracy on the graph.

Table 1: Dataset for task 2.

ID	Gender	Graduation	Eye Color	Class
1	F	College	brown	1
2	F	College	brown	1
3	F	College	gray	0
4	F	High School	blue	0
5	F	High School	blue	0
6	F	High School	brown	0
7	F	High School	gray	1
8	F	Middle School	blue	1
9	F	Middle School	blue	1
10	F	Middle School	brown	1
11	M	High School	blue	0
12	M	High School	brown	0
13	M	High School	brown	0
14	M	High School	gray	0
15	M	High School	green	0
16	M	High School	green	0
17	M	Middle School	brown	1
18	M	Middle School	gray	1
19	M	Middle School	green	1
20	M	Middle School	green	1

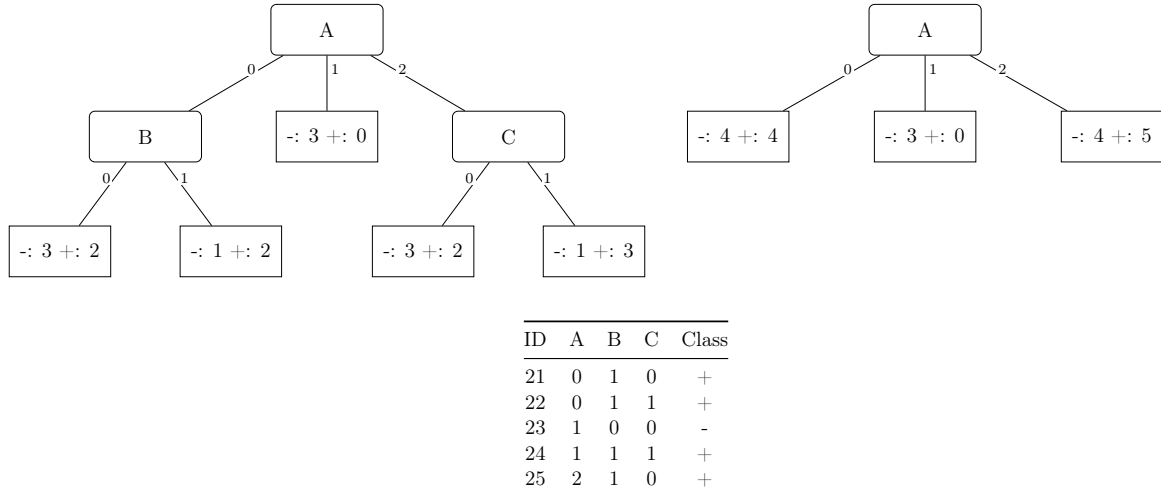


Figure 1: Two decision trees generated from the same training set and validation data.

Exercise 4

1. Create a confusion matrix from Table 2. Assume that “Y” serves as positive class. Calculate accuracy, precision, recall, F1-measure score, sensitivity and specificity.

Table 2: Actual and predicted labels for a classification problem.

	True Label	Predicted Label
1	N	N
2	N	N
3	N	Y
4	N	N
5	Y	N
6	N	Y
7	N	N
8	Y	Y
9	N	N
10	Y	Y

2. Consider a model that has an accuracy of $p = 85\%$ when evaluated on $N = 200$ test instances. What is the 90% confidence interval for its true accuracy? The formula for the calculation of the confidence interval is given as:

$$p \pm Z_{1-\alpha/2} \sqrt{\frac{p(1-p)}{N}}$$

Refer Table 1 to obtain Z values.

Table 1: Confidence limits for the normal distribution.

$\Pr[X < Z]$	0.70	0.80	0.90	0.95	0.98	0.99
Z	0.52	0.84	1.28	1.64	2.05	2.33

How would the confidence interval change if the size of the test set was 50 or 500, respectively?

- Summarize the performance of the classifiers from Table 3 to calculate the number of wins, draws and losses when comparing the classifiers.

To test whether the difference in performance between a classifier A and a classifier B is significant, consider the following approach. Let N be the size of a given dataset, p_A be the accuracy of classifier A , p_B be the accuracy of classifier B , and

$$p = \frac{p_A + p_B}{2}$$

be the average accuracy for both classifiers. To test whether classifier A is significantly better than B , the following Z -statistic is used:

$$Z = \frac{p_A - p_B}{\sqrt{\frac{2p(1-p)}{N}}}$$

Use a significance level of $\alpha = 5\%$ for the pairwise comparisons.
Hint: Classifier A wins if $Z > 1.64$.

Table 3:

Dataset	N	NB	DT
Congressional Voting Records	435	0.901	0.949
Dermatology	366	0.978	0.956
Employee attrition	1470	0.801	0.805

- An ensemble classifier is a committee of multiple base classifiers. To classify a test instance, the votes from all base classifiers are combined, e.g., using majority voting. Ensemble classifiers frequently outperform their base counterparts in terms of generalization performance. A random forest is an ensemble of decision trees. Explain how a random forest is induced. What are the “random” components of a random forest?
- Explain the following classifier evaluation methods: (a) hold-out method, (b) random subsampling, (c) cross-validation, (d) stratified cross-validation, (e) leave-one-out cross validation and (f) bootstrap.

Exercise 5

1. Evaluate the performance of two classification models, M_1 and M_2 . Table 1 shows the posterior probabilities for the positive class obtained by applying the models to the test set. Plot the ROC curve for both M_1 and M_2 and calculate the area under the ROC curve (AUC). Explain which model is better.
2. For model M_1 , suppose the cutoff threshold is set to be $t = 0.5$, i.e., any test instances whose posterior probability is greater than t will be classified as positive. Compute the precision, recall, and F1-measure for the model at this threshold value.
3. Repeat the analysis for task 2 using the same cutoff threshold on model M_2 . Compare the F1-measure results for both models. Which model is better? Are the results consistent with the conclusions drawn from the ROC curves in task 1?
4. Repeat task 2 for model M_1 using the threshold $t = 0.1$. Which threshold is better, $t = 0.5$ or $t = 0.1$? Are the results consistent with the conclusions drawn from the ROC curves in task 1?

Table 1: Posterior probabilities for the positive class obtained by two classifiers, M_1 and M_2 , on 10 test instances.

ID	True Class	$P(Y = + \mid X, M_1)$	$P(Y = + \mid X, M_2)$
1	—	0.35	0.80
2	—	0.05	0.40
3	+	0.70	0.10
4	+	0.45	0.25
5	+	0.40	0.05
6	—	0.30	0.15
7	+	0.80	0.60
8	—	0.15	0.20
9	+	0.65	0.45
10	—	0.60	0.30

5. Let the cost matrix in Table 2 be given. The total cost C of a model M can be calculated as:

$$C(M) = p \cdot (1 - \text{TPR}_M) \cdot c_{\text{FN}} + (1 - p) \cdot \text{FPR}_M \cdot c_{\text{FP}} + p \cdot \text{TPR}_M \cdot c_{\text{TP}} + (1 - p) \cdot (1 - \text{FPR}_M) \cdot c_{\text{TN}}.$$

Calculate the total cost for the models in Figure 1 for $p = 0.1$. Which model achieves the lowest cost?

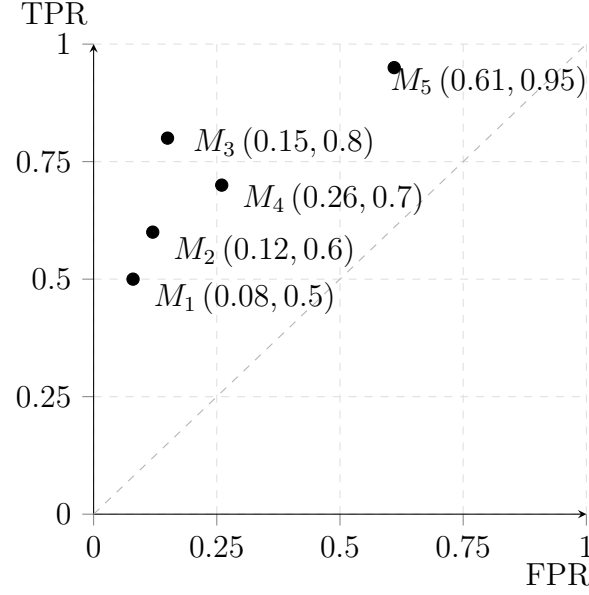


Figure 1: ROC points of five model candidates.

Table 2: Cost matrix.

	Classifier assignment	
	−	+
True class membership −	$c_{\text{TN}} = 0$	$c_{\text{FP}} = 2$
True class membership +	$c_{\text{FN}} = 10$	$c_{\text{TP}} = -1$

Exercise 6

1. What is (unsupervised) clustering? What is the difference between classification and clustering? Give examples from real life.
2. Describe the K-Means algorithm and its objective function. What is a cluster centroid? How is its position determined? How/When does the algorithm terminate? List the advantages and disadvantages of the algorithm.
3. The similarity/dissimilarity between objects can be described in different ways. Explain Manhattan distance, Euclidean distance and Cosine similarity. What are the formulas, value ranges and significance of these similarity functions?
4. Execute K-Means with $K = 2$ and Euclidean distance on the data below: assign all objects to a cluster, compute the SSE and find the centroid positions in each iteration. Instead of a random initialisation the initial centroids are p_{10} and p_{11} .

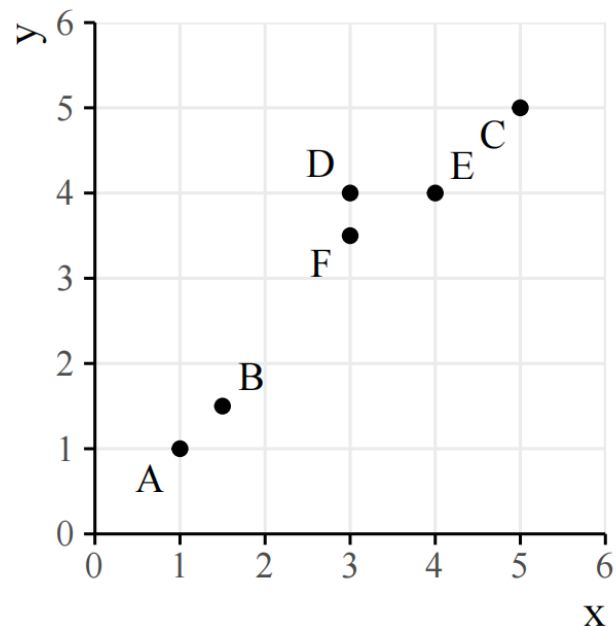
ID	x	y
p1	1.0	1.0
p2	2.0	2.0
p3	3.0	1.0
p4	3.0	2.5
p5	2.0	8.0
p6	2.0	6.0
p7	2.5	7.0
p8	4.0	7.0
p9	6.0	2.0
p10	7.0	3.0
p11	6.5	1.0

5. Explain the Bisecting K-means algorithm. How is it different from standard k-means? Explain how the process is carried out.

Exercise 8

1. Explain the difference between divisive and agglomerative hierarchical clustering.
2. Name the different inter-cluster distance measures (linkage method) and describe how they are computed.
3. Consider the dataset of 6 points given below. Perform an agglomerative clustering using Single/MIN linkage with Euclidean distance and create the dendrogram.

	x	y
A	1.0	1.0
B	1.5	1.5
C	5.0	5.0
D	3.0	4.0
E	4.0	4.0
F	3.0	3.5



4. Compute the distance between clusters $C_1 = \{A, B\}$ and $C_2 = \{D, E, F\}$ using:
 - (a) Complete/MAX linkage
 - (b) Group Average
 - (c) Ward's method
5. W.r.t your computations in question 4: If a new point $Z = (2, 2)$ was added to C_1 , the distance between C_1 and C_2 as computed by MAX linkage will:
 - Remain the same
 - not remain the same